NAME: _____ Geometry (FL B.E.S.T.)

9.6 Inscribed Triangles & Quadrilaterals

A polygon is **cyclic** when all of its vertices lie on a single circle. For a cyclic quadrilateral, opposite angles are always **supplementary** ($180°$ total) — this follows directly from the Inscribed Angle Theorem, since each pair of opposite angles intercepts a pair of arcs that together make up the entire $360°$ circle. The converse is also true and useful: if a quadrilateral's opposite angles are supplementary, it can be inscribed in a circle. **Thales' Theorem** (from the previous lesson) is the triangle version of this idea: an inscribed triangle with one side as a diameter always has a right angle opposite that diameter. This worksheet combines both ideas, including the proof of the cyclic-quadrilateral rule and problems that mix arcs, angles, and algebra.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Cyclic quadrilateral has $m\angle A = 78°$. Find $m\angle C$.

2. Cyclic quadrilateral has $m\angle B = 105°$. Find $m\angle D$.

3. $m\angle A = 5x$ and $m\angle C = (3x + 20)°$ (opposite). Find x and both angles.

4. Diameter $\overline{AB}$, point C on circle. $m\angle A = 50°$. Find $m\angle B$.

5. Opposite angles $110°$ and $70°$. Can it be cyclic?

6. Opposite angles $88°$ and $95°$. Can it be cyclic?

7. $m\angle B = (2x + 10)°$ and $m\angle D = (4x - 4)°$ (opposite). Find x and both angles.

8. Arcs of a cyclic quad: $70°$, $100°$, $85°$, and x. Find x.

9. In a cyclic quad, $\angle A$ intercepts arc $BCD = 160°$. Find $m\angle A$.

10. Inscribed right triangle; both acute angles equal. Find each acute angle.

11. $m\angle A = 125°$. Find $m\angle C$.

12. What must be true of opposite angles in a cyclic quadrilateral?

13. Diameter hypotenuse; one acute angle is $32°$. Find the other.

14. Cyclic quadrilateral: $m\angle A = (3x + 20)^\circ$ and $m\angle C = (2x + 10)^\circ$ (opposite). Find x, $m\angle A$, and $m\angle C$. Then, if $m\angle B = (4x - 15)^\circ$, find $m\angle D$.

15. In cyclic quadrilateral $ABCD$, diagonal $\overline{AC}$ is a diameter of the circle. Explain why both $m\angle ABC$ and $m\angle ADC$ must equal 90°.

✓ Answer Key

1. 102°
2. 75°
3. $x = 20$; angles 100°, 80°
4. 40°
5. Yes
6. No
7. $x = 29$; angles 68°, 112°
8. 105°
9. 80°
10. 45° each
11. 55°
12. They are supplementary (sum to 180°)
13. 58°
14. $x = 30$; $m\angle A = 110^\circ$, $m\angle C = 70^\circ$; $m\angle B = 105^\circ$, so $m\angle D = 75^\circ$
15. $\angle ABC = 90^\circ$ by Thales' Theorem; since $ABCD$ is cyclic, $\angle ADC = 180^\circ - 90^\circ = 90^\circ$ too